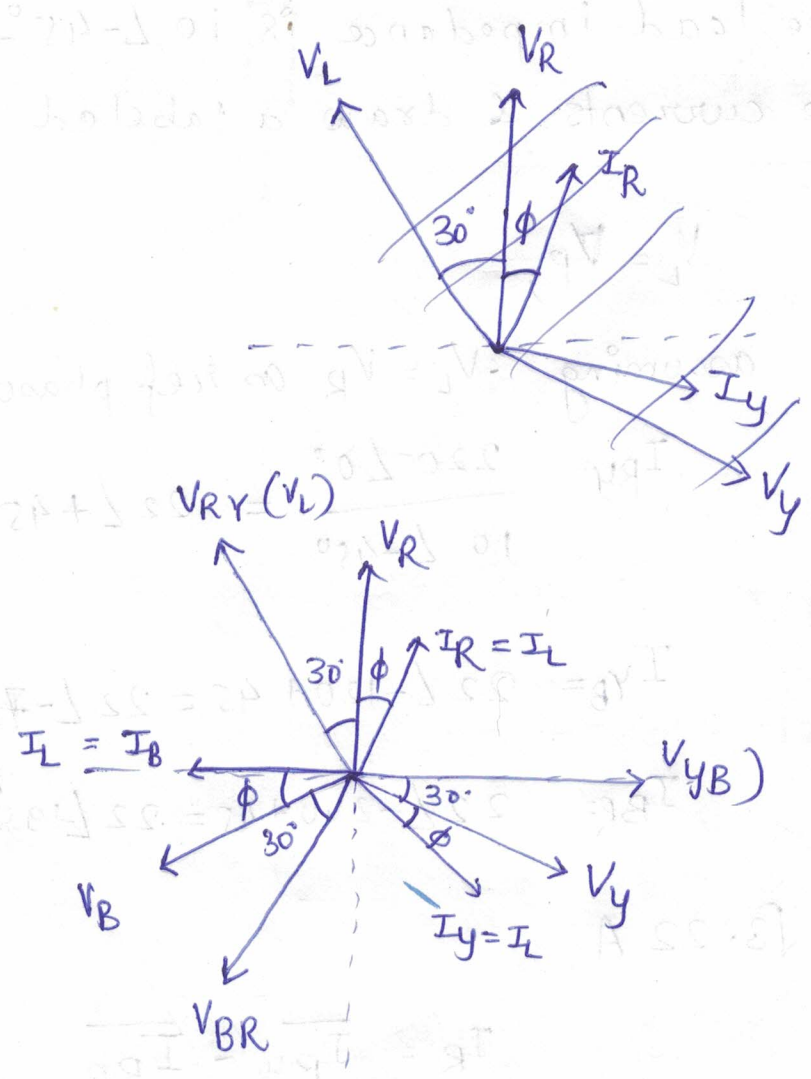


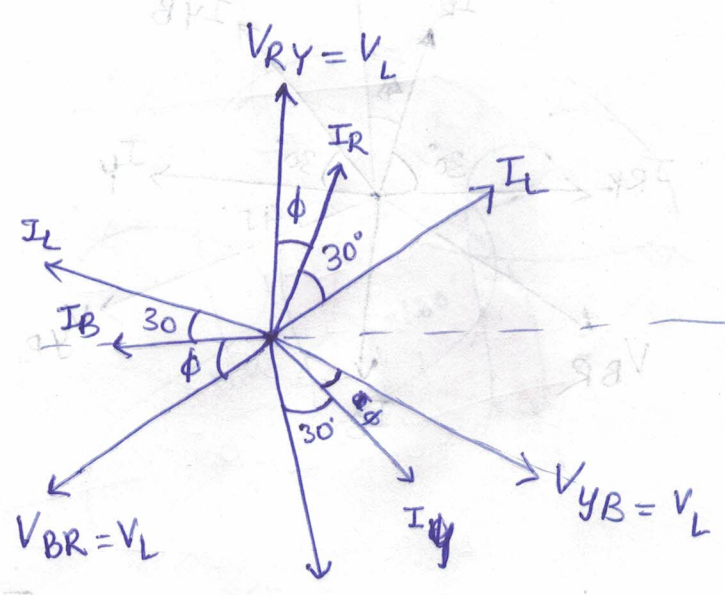
Phasor diagram for star connected load:-



* Line voltage in star connected ~~load~~ leads resp. phase voltage by 30° .

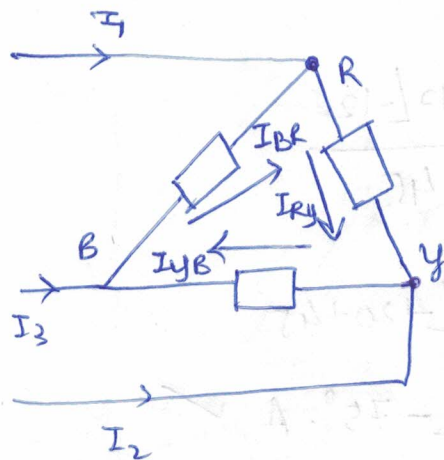
Complete Phasor diagram for star connected s/m.

• Line current in in delta load lags behind phase current by 30° .



Complete Phasor diagram for delta connected s/m.

A three phase 220V source feeds a delta connected 3 ϕ load where per phase load impedance is $10\angle -45^\circ \Omega$. Find RMS line current, phase currents. Draw labeled phasor diagram.



$$V_L = V_P$$

$$\therefore I_{RY} = \frac{V_R}{Z} = \frac{220}{10\angle -45^\circ} = 22\angle 45^\circ \text{ A}$$

$$I_{YB} = \frac{V_Y}{10\angle -45^\circ} = 22\angle -120^\circ + 45^\circ = 22\angle -75^\circ \text{ A}$$

$$I_{BR} = \frac{V_B}{10\angle -45^\circ} = 22\angle -240^\circ + 45^\circ = 22\angle -195^\circ \text{ A}$$

$$\bar{I}_1 = \bar{I}_{RY} - \bar{I}_{BR}$$

$$I_1 = 22\angle 45^\circ - 22\angle -195^\circ$$

$$= 22[\cos 45^\circ + j\sin 45^\circ - \cos 195^\circ + j\sin 195^\circ]$$

$$I_1 = 22\sqrt{3}\angle 15^\circ$$

$$I_2 = \bar{I}_{YB} - \bar{I}_{RY}$$

$$I_2 = 22(\angle -75^\circ - \angle 45^\circ)$$

$$= 22[\cos 75^\circ - j\sin 75^\circ - \cos 45^\circ - j\sin 45^\circ]$$

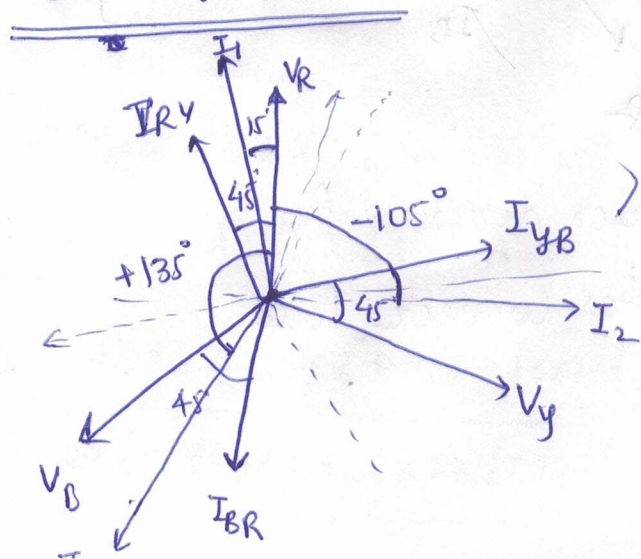
$$I_2 = 22\sqrt{3}\angle -105^\circ$$

$$I_3 = \bar{I}_{BR} - \bar{I}_{YB}$$

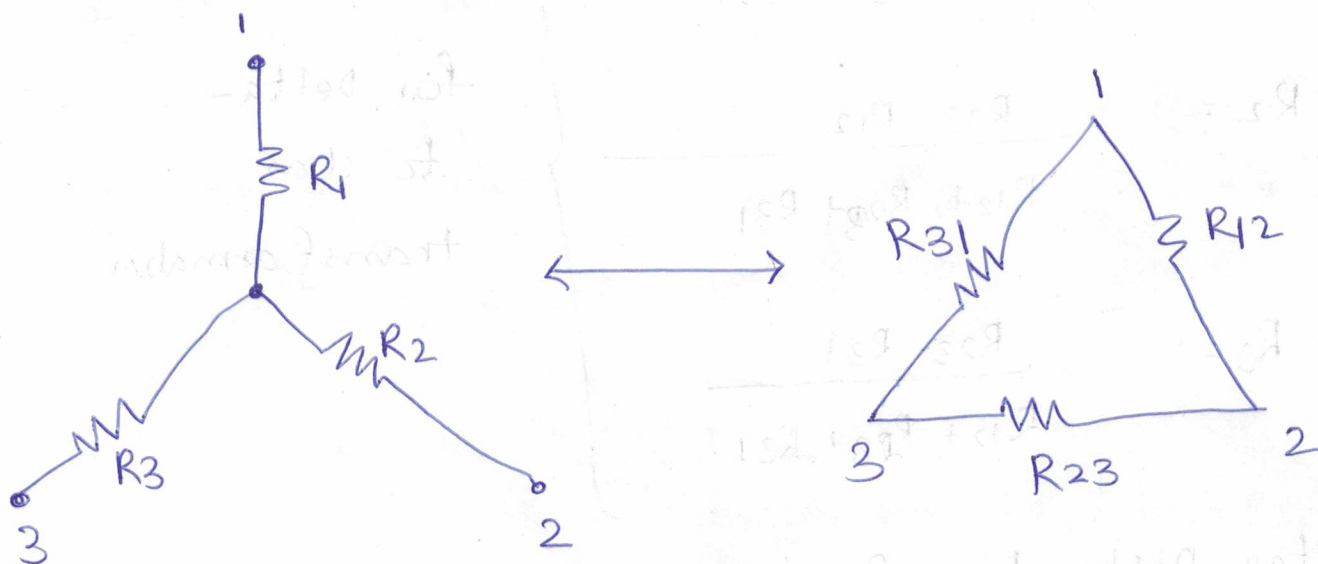
$$I_3 = 22\angle -195^\circ - 22\angle -75^\circ$$

$$= 22[\cos 195^\circ - j\sin 195^\circ - \cos 75^\circ + j\sin 75^\circ]$$

$$I_3 = 22\sqrt{3}\angle 135^\circ$$



Star-Delta Conversion :-



For the two ckt's to be equivalent, the impedance or resistance betn any two terminals must be equal in both cases.

$$\text{fr } 1, 2 \quad R_1 + R_2 = (R_{23} + R_{31}) \parallel R_{12}$$

$$R_1 + R_2 = \frac{(R_{23} + R_{31}) \times R_{12}}{R_{12} + R_{23} + R_{31}} \quad \text{--- (1)}$$

$$R_2 + R_3 = \frac{(R_{31} + R_{12}) \times R_{23}}{R_{12} + R_{23} + R_{31}} \quad \text{--- (2)}$$

$$\& R_1 + R_3 = \frac{(R_{23} + R_{12}) \times R_{31}}{R_{12} + R_{23} + R_{31}} \quad \text{--- (3)}$$

$$R_1 + R_2 + R_3 = \frac{R_{12} \cdot R_{23} + R_{12} \cdot R_{31} + R_{23} \cdot R_{31}}{R_{12} + R_{23} + R_{31}}$$

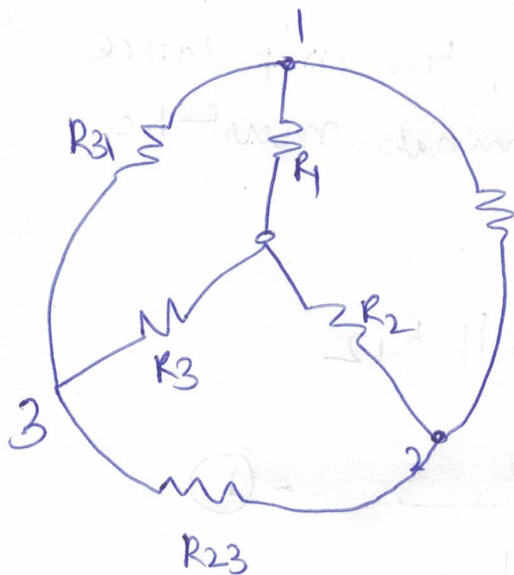
$$\therefore R_1 = \frac{R_{12} \cdot R_{31}}{R_{12} + R_{23} + R_{31}}$$

$$R_2 = \frac{R_{23} \cdot R_{12}}{R_{12} + R_{23} + R_{31}}$$

$$R_3 = \frac{R_{23} \cdot R_{31}}{R_{12} + R_{23} + R_{31}}$$

for Delta-
to star
transformation.

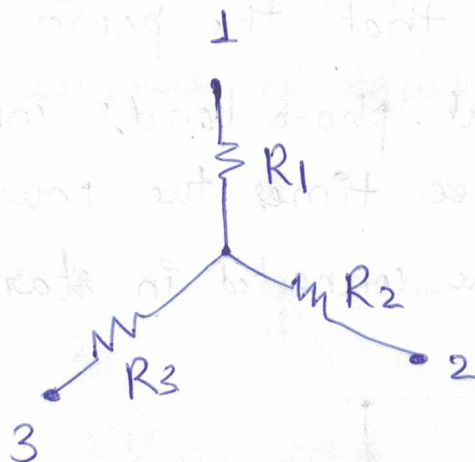
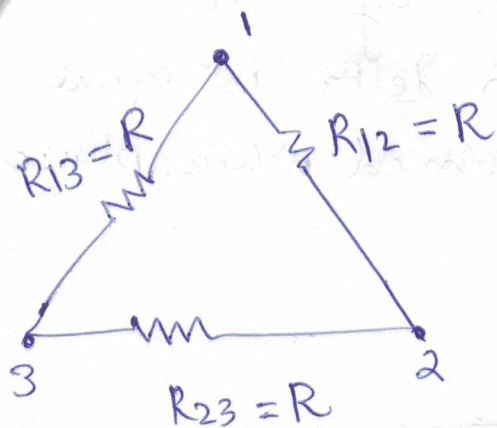
for Star-Delta transformation - [Star to Delta]



$$R_{12} = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3}$$

$$R_{23} = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1}$$

$$R_{31} = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2}$$

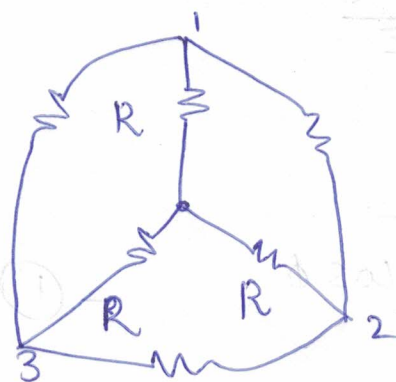


$$R_1 = \frac{R_{13} \cdot R_{12}}{R_{12} \cdot R_{23} + R_{23} \cdot R_{31} + R_{13} \cdot R_{23}} [R_{12} + R_{31} + R_{23}]$$

$$= \frac{R^2}{3R} = \frac{R}{3}$$

$$\underline{R_2 = R/3}, \quad \underline{R_3 = R/3}$$

* If R in Δ network is equal, then in Y network the equivalent impedance is $1/3$ times.



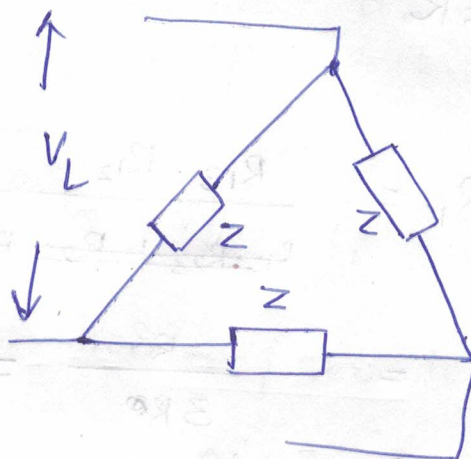
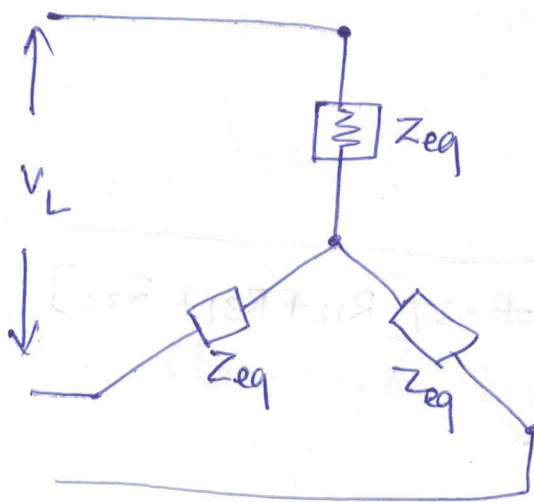
$$R_{12} = \frac{R^2 + R^2 + R^2}{R} = 3R$$

$$R_{23} = 3R$$

$$R_{31} = 3R$$

* If R in Y network is equal, then in Δ network, the equivalent impedance/resistance is 3 times.

* Show that the power consumed by three identical phase loads connected in delta is equal to three times the power consumed when phase loads are connected in star.



for delta connected load,

$$V_p = V_L$$

$$I_p = \frac{V_L}{Z}, \quad I_L = \sqrt{3} I_p = \frac{\sqrt{3} V_L}{Z}$$

$$P_T = \sqrt{3} V_L I_L \cos \phi.$$

$$= \sqrt{3} V_L \cdot \frac{(\sqrt{3} V_L)}{Z} \cdot \cos \phi = \frac{3 V_L V_L \cos \phi}{Z} \quad \text{--- (1)}$$

for star connected N/w -

$$Z_{eq} = \frac{Z^2}{Z+Z+Z} = \frac{Z}{3}$$

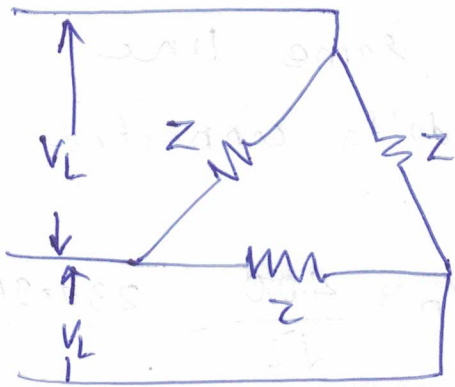
$$V_p = \frac{V_L}{\sqrt{3}}$$

$$I_p = \frac{V_L}{\sqrt{3} Z} = I_L$$

$$P_s = 3 V_p I_p \cos \phi = \sqrt{3} V_L I_L \cos \phi = \frac{\sqrt{3} V_L V_L \cos \phi}{\sqrt{3}}$$

$$\therefore \boxed{P_\Delta = 3 P_s}$$

Show that power consumed in ~~in~~ delta connected balanced load is same as that of equivalent star connected load.

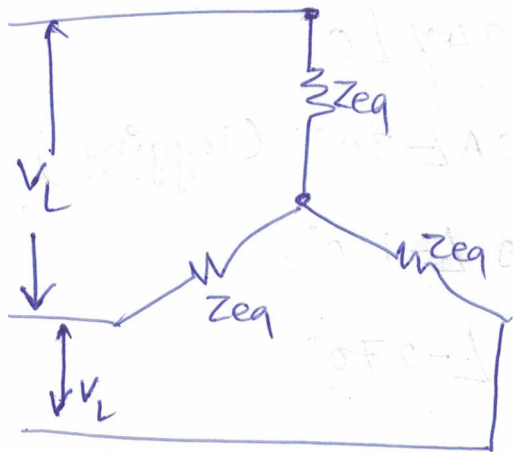


$$I_p = \frac{V_L}{Z}, \quad |I_L| = \sqrt{3} |I_p|$$

$$P_T = \sqrt{3} V_L I_L \cos \phi$$

$$= 3 V_p I_p \cos \phi$$

$$P_T = \frac{3 V_L V_L}{Z} \cos \phi = \frac{3 V_L^2}{Z} \cos \phi$$



$$Z_{eq} = \frac{Z^2}{Z+Z+Z} = \frac{Z}{3}$$

$$V_p = \frac{V_L}{\sqrt{3}}, \quad I_p = \frac{V_L}{\sqrt{3} \left(\frac{Z}{3} \right)} = \frac{\sqrt{3} V_L}{Z}$$

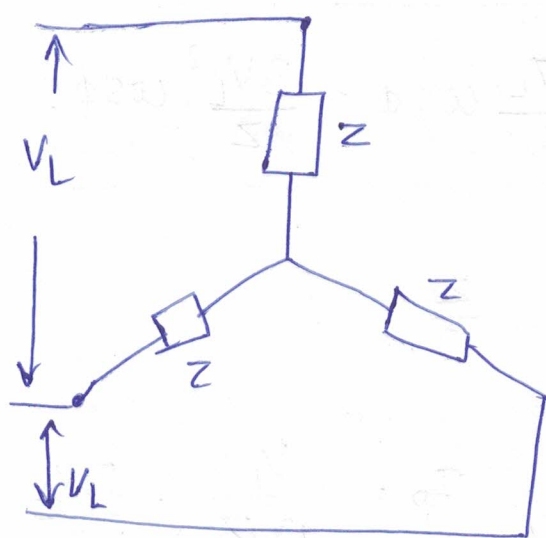
$$P_T = 3 V_p I_p \cos \phi$$

$$= \frac{3 V_L}{\sqrt{3}} \times \frac{\sqrt{3} V_L}{Z} \cos \phi$$

$$P_T = \frac{3 V_L^2}{Z} \cos \phi$$

$I_{L\Delta} = \frac{\sqrt{3} \cdot V_p}{Z}$, $I_{LY} = \frac{\sqrt{3} \cdot V_L}{Z}$ line current is same for both the network.

* A balanced star connected load is fed from a three phase 50Hz, 400V supply. The per-phase impedance is $4.62 \angle 30^\circ \Omega$. Calculate line current magnitude. Find equivalent Δ representation of load & show that same line current is drawn by equivalent delta connected load.



$$V_L = 400V, \quad V_P = \frac{400}{\sqrt{3}} = 230.94V$$

$$I_P = \frac{230.94}{4.62 \angle 30^\circ} = 49.99A$$

$$V_R = 230.94V \angle 0^\circ$$

$$I_R = 49.99A \angle -30^\circ \text{ (lagging)}$$

$$I_Y = 49.99A \angle -150^\circ$$

$$I_B = 49.99A \angle -270^\circ$$

$$|I_L| = |I_R| = |I_Y| = |I_B|$$

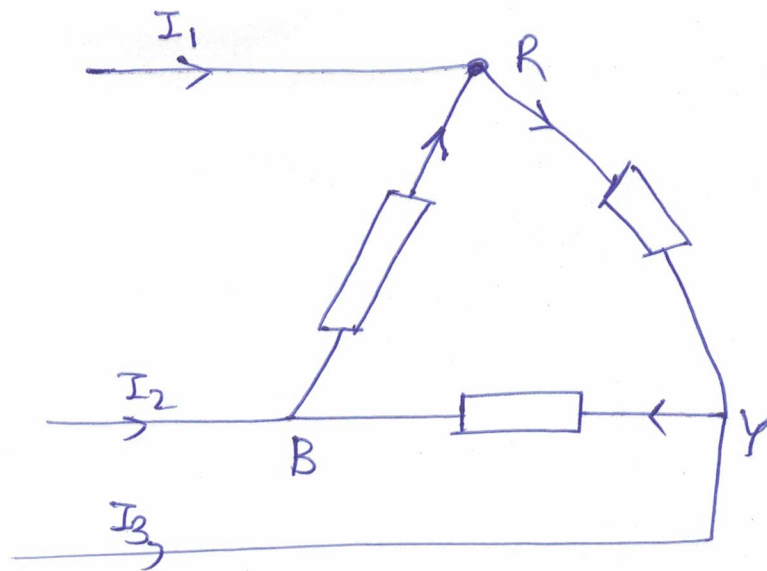
in delta s/m -

$$Z_{eq} = 3Z = 3 \times 4.62 \angle 30^\circ = 13.86 \angle 30^\circ \Omega$$

$$I_P = \frac{V_L}{Z} = \frac{400}{13.86 \angle 30^\circ} = 28.86A \angle -30^\circ$$

$$I_R = 28.86A \angle -30^\circ, \quad I_Y = 28.86A \angle -150^\circ,$$

$$I_B = 28.86A \angle -270^\circ$$



$$I_1 = I_R - I_B$$

$$= 28.86 \angle 30^\circ - 28.86 \angle -270^\circ$$

$$= 28.86 [\cos 30 - j \sin 30 - \cos 270 + j \sin 270]$$

$$= 28.86 [0.866 - j 1.5]$$

$$I_1 = 28.86 \sqrt{3} \angle -60^\circ \text{ A}$$

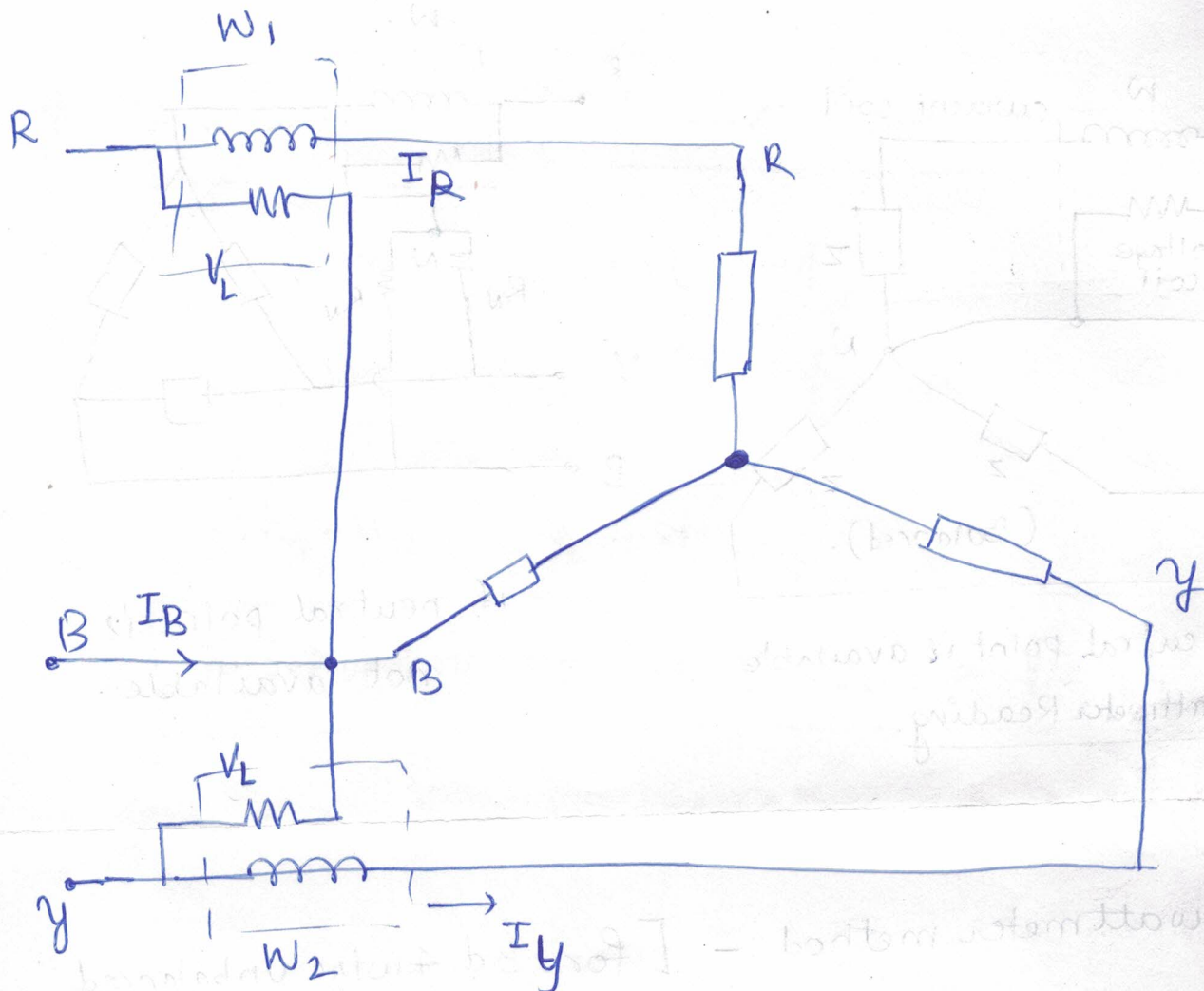
$$\approx 49.99 \text{ A}$$

$$I_2 = 28.86 \times \sqrt{3} \angle -180^\circ \text{ A}$$

$$I_3 = 28.86 \sqrt{3} \angle -300^\circ \text{ A}$$

* Two wattmeter method:- 3ϕ , 3 wire s/m

→ Both balanced, unbalanced,



In 3ϕ 3 wire s/m, sum of line currents is zero.

$$i_R + i_Y + i_B = 0 \quad [\text{balanced load}]$$

or unbalanced

Total instantaneous power = $V_R \cdot i_R + V_Y i_Y + V_B \cdot i_B$.

$$i_B = -(i_R + i_Y)$$

$$P_t = V_R i_R + V_Y i_Y + V_B (i_R + i_Y)$$

$$= i_R (V_R - V_B) + i_Y (V_Y - V_B)$$

Instantaneous power of W_1

Instantaneous power of W_2

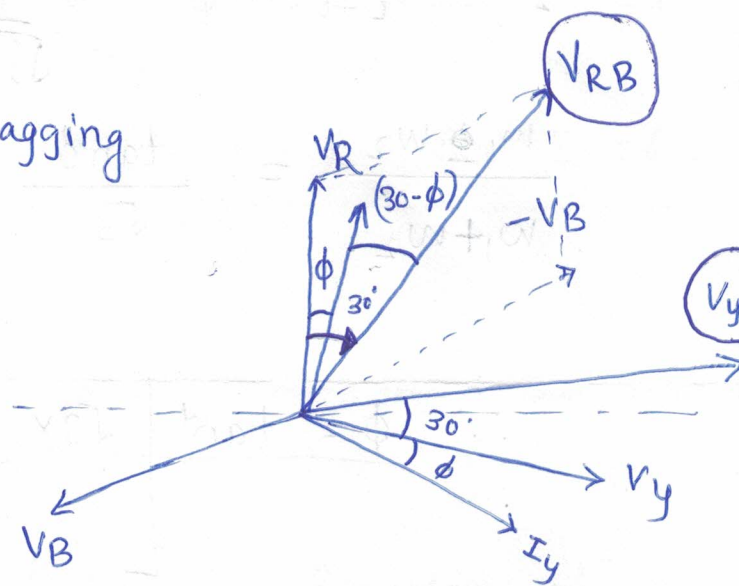
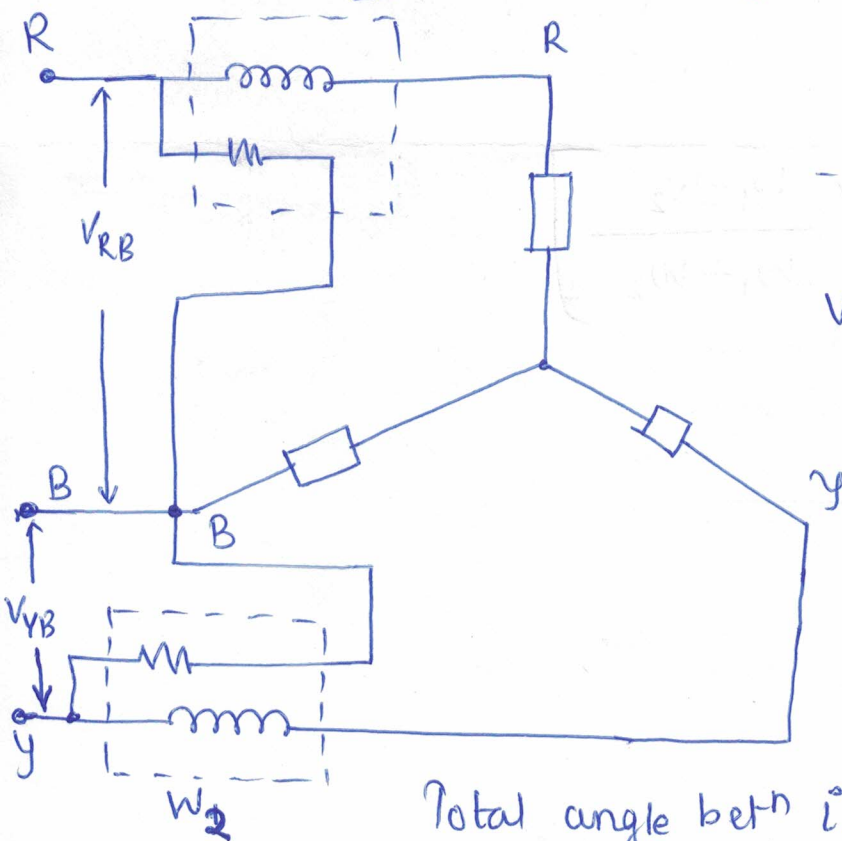
$$= i_R V_{RL(RB)} + i_Y (V_{YB})$$

Total instantaneous power = $P_1 + P_2$

$$P_{avg} = P_1 + P_2$$

* In some condition, wattmeter may give zero reading even when load is connected, under this case either the connections of either voltage coil or current coil needs to be reversed. The reading so ~~data~~ obtained must be considered -ve for computing total power.

* Phasor diagram:- [assuming lagging load]



$$\begin{aligned} V_{RB} &= \bar{V}_R - \bar{V}_B \\ &= V_R \angle 0^\circ - V_B \angle -120^\circ \\ &= \sqrt{3} V_R \angle -30^\circ \end{aligned}$$

$$V_{YB} = \sqrt{3} V_Y \angle 30^\circ$$

Total angle betⁿ $\bar{I}$ & $\bar{V}$ for Y-B - $(30^\circ + \phi)$
 ||
 for R-B - $(30^\circ - \phi)$

$$\therefore W_1 = \cancel{V_L} V_L I_L \cos(30^\circ - \phi)$$

$$I_L = I_R$$

$$W_2 = V_L I_L \cos(30^\circ + \phi)$$

$$I_L = I_Y$$

$$\cos(30^\circ - \phi) > \cos(30^\circ + \phi)$$

$$W_1 + W_2 = V_L I_L [\cos(30^\circ - \phi) + \cos(30^\circ + \phi)]$$

$$= V_L I_L [\cos 30^\circ \cdot \cos \phi + \sin 30^\circ \sin \phi + \cos 30^\circ \cos \phi + \sin 30^\circ \sin \phi]$$

$$= 2 V_L I_L \cos 30^\circ \cos \phi$$

$$= \frac{2 V_L I_L \sqrt{3}}{2} \cos \phi = \underline{\underline{\sqrt{3} \cdot V_L I_L \cos \phi = P_T \cdot W}}$$

$$W_1 - W_2 = V_L I_L \sin \phi = \frac{1}{\sqrt{3}} Q_T \quad \therefore Q_T = \sqrt{3} (W_1 - W_2)$$

$$\frac{W_1 - W_2}{W_1 + W_2} = \frac{\tan \phi}{\sqrt{3}}$$

$$\therefore \phi = \tan^{-1} \left[\sqrt{3} \times \frac{W_1 - W_2}{W_1 + W_2} \right]$$

$$\boxed{P.f = \cos \phi}$$

Leading load:-

$$W_1 = V_L I_L \cos(30 + \phi)$$

$$W_2 = V_L I_L \cos(30 - \phi)$$

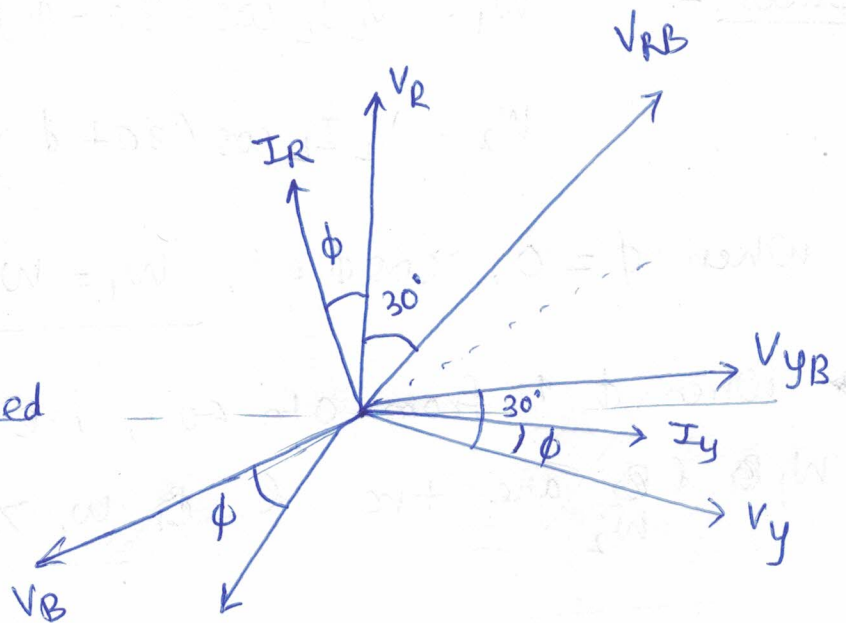
Readings are interchanged

$$W_1 + W_2 = \sqrt{3} V_L I_L \cos \phi$$

but

$$W_1 - W_2 = -V_L I_L \sin \phi$$

$$W_2 - W_1 = V_L I_L \sin \phi$$



$$\cos(30 + \phi) < \cos(30 - \phi)$$

In general,

$$\tan \phi = \frac{\sqrt{3} \times [\text{greater wattmeter reading} - \text{smaller wattmeter reading}]}{\text{sum of wattmeter reading}}$$